

Advanced Computer Networks (ACN) Lab Report

Title: Reliable File Transfer over UDP using Stop-and-Wait ARQ Protocol

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1. Objective

The Goal of this experiment is to implement a file transfer mechanism using UDP using Stop and wait ARQ mechanism

With main features as –

- Acknowledgments (ACKs)
 - Timers and retransmissions
 - Sequence numbers (0/1 toggle)
 - Handling packet loss using Linux tc to simulate network unreliability
-

2. Screenshots

1. When packet loss is 0% client side + server side

```
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
EOF received. File transfer complete.

=====
Transfer Summary: Server Side
=====
File saved as: Reciecd_files/received_file.pdf
Simulated packet loss: 0.0%
File size received: 2208728 bytes
Total packets received: 2157
Duplicates: 0
Dropped (simulated): 0
Total time: 0.56 seconds
=====
File 'Reciecd_files/received_file.pdf' written successfully.
PS C:\Users\atish\Desktop\UDP_Assignment> █

Sent packet seq=0 (1024 bytes)
ACK 0 received

Sent packet seq=1 (1024 bytes)
ACK 1 received

Sent packet seq=0 (1024 bytes)
ACK 0 received

Sent packet seq=1 (1024 bytes)
ACK 1 received

Sent packet seq=0 (984 bytes)
ACK 0 received

=====
Transfer Summary - Client Side
=====
File sent successfully.
File name: input_file.pdf
File size: 2208728 bytes
Total packets sent: 2157
Retransmissions: 0
Total transfer time: 0.56 seconds
=====
PS C:\Users\atish\Desktop\UDP_Assignment> █
```

2. When packet loss is 10% client side + server side

```
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
EOF received. File transfer complete.
```

Transfer Summary: Server Side

```
File saved as: Reciecd_files/received_file.pdf
Simulated packet loss: 10.0%
File size received: 2208728 bytes
Total packets received: 2157
Duplicates: 0
Dropped (simulated): 240
Total time: 4.39 seconds
```

```
File 'Reciecd_files/received_file.pdf' written successfully.
```

```
PS C:\Users\atish\Desktop\UDP_Assignment> █
```

```
Sent packet seq=0 (1024 bytes)
ACK 0 received
```

```
Sent packet seq=1 (1024 bytes)
ACK 1 received
```

```
Sent packet seq=0 (1024 bytes)
ACK 0 received
```

```
Sent packet seq=1 (1024 bytes)
ACK 1 received
```

```
Sent packet seq=0 (984 bytes)
ACK 0 received
```

Transfer Summary - Client Side

```
File sent successfully.
File name: input_file.pdf
File size: 2208728 bytes
Total packets sent: 2157
Retransmissions: 240
Total transfer time: 4.39 seconds
```

```
PS C:\Users\atish\Desktop\UDP_Assignment> █
```

3. When packet loss is 20% client side + server side

```
Received seq=0, sent ACK 0
Simulated packet loss (dropping packet)
Received seq=1, sent ACK 1
Simulated packet loss (dropping packet)
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Simulated packet loss (dropping packet)
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Simulated packet loss (dropping packet)
Simulated packet loss (dropping packet)
Received seq=0, sent ACK 0
EOF received. File transfer complete.

=====
                        Transfer Summary: Server Side
=====
File saved as: Recieced_files/received_file.pdf
Simulated packet loss: 20.0%
File size received: 2208728 bytes
Total packets received: 2157
Duplicates: 0
Dropped (simulated): 549
Total time: 8.78 seconds
=====
File 'Recieced_files/received_file.pdf' written successfully.
PS C:\Users\atish\Desktop\UDP_Assignment>

Sent packet seq=0 (1024 bytes)
Timeout waiting for ACK. Resending seq=0
Sent packet seq=0 (1024 bytes)
ACK 0 received

Sent packet seq=1 (1024 bytes)
ACK 1 received

Sent packet seq=0 (984 bytes)
Timeout waiting for ACK. Resending seq=0
Sent packet seq=0 (984 bytes)
Timeout waiting for ACK. Resending seq=0
Sent packet seq=0 (984 bytes)
ACK 0 received

=====
                        Transfer Summary - Client Side
=====
File sent successfully.
File name: input_file.pdf
File size: 2208728 bytes
Total packets sent: 2157
Retransmissions: 549
Total transfer time: 8.78 seconds
=====
PS C:\Users\atish\Desktop\UDP_Assignment>
```

4. When packet loss is 30% client side + server side

```
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Simulated packet loss (dropping packet)
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
Received seq=1, sent ACK 1
Received seq=0, sent ACK 0
EOF received. File transfer complete.

=====
                        Transfer Summary: Server Side
=====
File saved as: Recieced_files/received_file.pdf
Simulated packet loss: 30.0%
File size received: 2208728 bytes
Total packets received: 2157
Duplicates: 0
Dropped (simulated): 924
Total time: 14.51 seconds
=====
File 'Recieced_files/received_file.pdf' written successfully.
PS C:\Users\atish\Desktop\UDP_Assignment> █

ACK 0 received

Sent packet seq=1 (1024 bytes)
Timeout waiting for ACK. Resending seq=1
Sent packet seq=1 (1024 bytes)
ACK 1 received

Sent packet seq=0 (1024 bytes)
ACK 0 received

Sent packet seq=1 (1024 bytes)
ACK 1 received

Sent packet seq=0 (984 bytes)
ACK 0 received

=====
                        Transfer Summary - Client Side
=====
File sent successfully.
File name: input_file.pdf
File size: 2208728 bytes
Total packets sent: 2157
Retransmissions: 924
Total transfer time: 14.51 seconds
=====
PS C:\Users\atish\Desktop\UDP_Assignment> █
```

3. Experimental Setup

Parameter	Description
OS	Windows 11
Programming Language	Python 3
Protocol	UDP
Timeout	0.1 second
Packet Size	1024 bytes
File Transferred	Input_file.pdf of 2157 KB size

4. Observations

File Used: input_file.pdf

File Size: 2157KB

Packet Loss	Transfer Time (s)	Retransmissions	File Size
0%	0.56 seconds	0	2157 KB
10%	4.39 seconds	240	2157 KB
20%	8.78 seconds	549	2157 KB
30%	14.51 seconds	924	2157 KB

5. Results and Analysis

1. At 0% packet loss, the transfer is smooth — no retransmissions.
 2. With increasing packet loss rates, the transfer time and retransmissions increase significantly.
 3. Stop-and-Wait ARQ ensures reliable delivery but becomes inefficient for high latency or lossy networks due to its waiting nature.
 4. Each lost packet or ACK triggers a retransmission, which increases total transfer time.
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6. Conclusion

- The experiment successfully demonstrated reliable file transfer over UDP using the Stop-and-Wait ARQ protocol.
 - The protocol's efficiency decreased with higher packet loss due to frequent retransmissions.
 - This highlights the trade-off between reliability and performance in lossy networks.
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7. Deliverables

File	Description
client_udp.py	Client-side implementation
server_udp.py	Server-side implementation
Report.pdf	This document with results and screenshots
README.md	Instructions to compile, run, and test the programs
Input_file.pdf	Test file used for transfer
submission.zip	Final zip file containing all above files
